

Netflix Content Insights: A Data-Driven Exploration Using SQL

Overview

This project applies advanced SQL queries to analyze Netflix's global content catalog with the goal of uncovering insights that can drive business growth. It explores viewing trends, genre popularity, regional production strengths, and actor/director impact to help media platforms and content teams make informed, data-driven decisions.

💡 Key Business Insights:

- **Content Strategy:** Understand viewer preferences (e.g., movies vs TV shows, top genres, most-watched countries).
- **Talent Impact:** Identify high-performing actors and directors to guide casting and partnerships.
- **Regional Trends:** Analyze country-specific content growth to support localization and expansion.
- **Catalog Optimization:** Spot outdated or underperforming content and recommend fresh additions.
- **User Safety:** Flag violent or sensitive content for better parental controls and personalized filtering.

Insights:

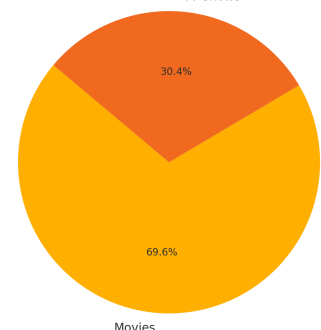
1. Distribution of Content Types: Movies vs TV Shows on Netflix

```
SELECT * FROM netflix_titles;

SELECT
  type AS Content_Type,
  COUNT(*) AS Total_Count
FROM
  netflix_titles
GROUP BY
  type
ORDER BY
  Total_Count DESC;
```

	Content_Type	Total_Count
1	Movie	6131
2	TV Show	2676

Distribution of Movies vs. TV Shows on Netflix



2. Most Common Content Rating by Type (Movies vs TV Shows)

```
SELECT TOP 1 WITH TIES
    type,
    rating AS most_frequent_rating,
    COUNT(*)
FROM netflix_titles
GROUP BY type, rating
ORDER BY
    RANK() OVER (PARTITION BY type ORDER BY COUNT(*) DESC);
```

100 %

Results Messages

	type	most_frequent_rating	(No column name)
1	Movie	TV-MA	2062
2	TV Show	TV-MA	1145

3. List of Movies Released in the Year 2021

```
SELECT
    title,
    type,
    release_year,
    rating,
    duration,
    listed_in
FROM netflix_titles
WHERE
    release_year = 2021 AND
    type = 'Movie'
ORDER BY
    title ASC;
```

100 %

Results Messages

	title	type	release_year	rating	duration	listed_in
219	The Ice Road	Movie	2021	PG-13	109 min	Action & Adventure
220	The Kissing Booth 3	Movie	2021	TV-14	114 min	Comedies, Romantic Movies
221	The Last Kids on Earth: Happ...	Movie	2021	TV-Y7	92 min	Children & Family Movies, Comedies
222	The Last Letter From Your Lo...	Movie	2021	TV-MA	111 min	Dramas, Romantic Movies
223	The Last Mercenary	Movie	2021	TV-MA	112 min	Action & Adventure, Comedies, International M...
224	The Loud House Movie	Movie	2021	TV-Y7	88 min	Children & Family Movies, Comedies, Music & ...
225	The Minimalists: Less Is Now	Movie	2021	TV-14	54 min	Documentaries
226	The Misadventures of Hedi a...	Movie	2021	TV-MA	99 min	Comedies, International Movies
227	The Mitchells vs. The Machines	Movie	2021	PG	114 min	Children & Family Movies, Comedies
228	The Paper Tigers	Movie	2021	PG-13	111 min	Action & Adventure, Comedies
229	The Rescue Jump	Movie	2021	TV-14	82 min	Documentaries

Query executed successfully.

4. Top 5 Countries with the Highest Volume of Netflix Content

```
74
75
76 SELECT
77     UNNEST(STRING_TO_ARRAY(country, ',')) as new_country,
78     COUNT(show_id) as total_content
79 FROM netflix
80 GROUP BY 1
81 ORDER BY 2 DESC
82 LIMIT 5
83
84
```

Data Output Messages Notifications

	new_country text	total_content bigint
1	United States	3211
2	India	1008
3	United Kingdo...	628
4	United States	479
5	Canada	271

5. Identifying the Longest Movie on Netflix by Duration

```
86
87
88 SELECT * FROM netflix
89 WHERE
90     type = 'Movie'
91     AND
92     duration = (SELECT MAX(duration) FROM netflix)
93
```

Data Output Messages Notifications

	max text
1	99 min

	show_id character varying (6)	type character varying (10)	title character varying (150)	director character varying (208)
1	s52	Movie	InuYasha the Movie 2: The Castle Beyond the Looking Glass	Toshiya Shinohara
2	s53	Movie	InuYasha the Movie 3: Swords of an Honorable Ruler	Toshiya Shinohara
3	s120	Movie	Here and There	JP Habac
4	s338	Movie	Good Luck Chuck	Mark Helfrich
5	s345	Movie	My Girl 2	Howard Zieff
6	s427	Movie	Cousins	Ainsley Gardiner, Briar Grace-Smith

6. Netflix Content Added in the Last 5 Years

```
SELECT
  *
FROM netflix
WHERE
  TO_DATE(date_added, 'Month DD, YYYY') >= CURRENT_DATE - INTERVAL '5 years'
```

7. All Movies and TV Shows Directed by Rajiv Chilaka

112	
113	SELECT * FROM netflix
114	WHERE director ILIKE '%Rajiv Chilaka%'
115	
116	
117	

	show_id character varying (6)	type character varying (10)	title character varying (150)	director character varying (208)
1	s407	Movie	Chhota Bheem - Neeli Pahaadi	Rajiv Chilaka
2	s408	Movie	Chhota Bheem & Ganesh	Rajiv Chilaka
3	s409	Movie	Chhota Bheem & Krishna: Mayanagari	Rajiv Chilaka
4	s410	Movie	Chhota Bheem & Krishna: Pataliputra- City of the Dead	Rajiv Chilaka
5	s411	Movie	Chhota Bheem And The Broken Amulet	Rajiv Chilaka
6	s412	Movie	Chhota Bheem And The Crown of Valhalla	Rajiv Chilaka
7	s413	Movie	Chhota Bheem and the Incan Adventure	Rajiv Chilaka
8	s414	Movie	Chhota Bheem and The ShiNobi Secret	Rajiv Chilaka

8. TV Shows with More Than 5 Seasons on Netflix

```
119
120 SELECT
121 *
122 FROM netflix
123 WHERE
124     type = 'TV Show'
125     AND
126     SPLIT_PART(duration, ' ', 1)::numeric > 5
```

	country character varying (150)	date_added character varying (50)	release_year integer	rating character varying (10)	duration character varying (15)
1	United Kingdom	September 24, 2021	2021	TV-14	9 Seasons
2	United States	September 15, 2021	2021	TV-PG	6 Seasons
3	United Kingdom	September 15, 2021	2021	TV-Y	6 Seasons
4	United States	September 15, 2021	1994	TV-PG	9 Seasons
5	United States	September 10, 2021	2021	TV-14	6 Seasons
6	United States	August 13, 2021	2021	TV-MA	7 Seasons
7	United States	August 1, 2021	2012	TV-14	7 Seasons
8	Japan	August 1, 2021	2014	TV-14	6 Seasons
9	United States	July 28, 2021	2021	TV-14	7 Seasons
10	United States	July 26, 2021	2019	TV-MA	10 Seasons
11	United States, Canada	July 6, 2021	2018	TV-Y	8 Seasons

9. Counting Number of Titles by Genre (Across All Content Types)

```
134
135 SELECT
136     UNNEST(STRING_TO_ARRAY(listed_in, ',')) as genre,
137     COUNT(show_id) as total_content
138 FROM netflix
139 GROUP BY 1
140
```

Data Output Messages Notifications

	genre text	total_content bigint
1	International TV Shows	774
2	Stand-Up Comedy	9
3	Spanish-Language TV Shows	172
4	Romantic Movies	613
5	Anime Features	21
6	TV Horror	11
7	Stand-Up Comedy & Talk Shows	34
8	Kids' TV	63

10. Yearly Contribution of Indian Content to Netflix – Top Performing Years

```
147
148 SELECT
149     EXTRACT(YEAR FROM TO_DATE(date_added, 'Month DD, YYYY')) as year,
150     COUNT(*) as yearly_content,
151     ROUND(
152         COUNT(*)::numeric/(SELECT COUNT(*) FROM netflix WHERE country = 'India')::numeric * 100
153     ,2)as avg_content_per_year
154 FROM netflix
155 WHERE country = 'India'
156 GROUP BY 1
157
```

	year	yearly_content	avg_content_per_year
	numeric	bigint	numeric
1	2018	333	34.26
2	2016	10	1.03
3	2019	203	20.88
4	2021	95	9.77
5	2020	189	19.44
6	2017	142	14.61

11. List of All Netflix Movies Categorized as Documentaries

```
162 SELECT * FROM netflix
163 WHERE
164     listed_in ILIKE '%documentaries%'
165
```

12. Content Without a Listed Director on Netflix

```
169
170 SELECT * FROM netflix
171 WHERE
172     director IS NULL
173
```

	show_id	type	title	director
	character varying (6)	character varying (10)	character varying (150)	
1	s2	TV Show	Blood & Water	[null]
2	s4	TV Show	Jailbirds New Orleans	[null]
3	s5	TV Show	Kota Factory	[null]
4	s11	TV Show	Vendetta: Truth, Lies and The Mafia	[null]
5	s15	TV Show	Crime Stories: India Detectives	[null]
6	s16	TV Show	Dear White People	[null]
7	s369	TV Show	Khawatr	[null]
8	s18	TV Show	Falsa identidad	[null]
9	s20	TV Show	Jaguar	[null]

13. Number of Movies Featuring Salman Khan Released in the Last 10 Years

```
178
179 SELECT * FROM netflix
180 WHERE
181     casts ILIKE '%Salman Khan%'
182     AND
183     release_year > EXTRACT(YEAR FROM CURRENT_DATE) - 10
184
185
186 14. Find the top 10 actors who have appeared in the highest number of movies produced.
```

Data Output Messages Notifications

		country character varying (150)	date_added character varying (50)	release_year integer	rating character varying (10)	duration character va
1	ndey, Sucheta Kh...	India	June 25, 2020	2015	TV-14	164 min
2		India	March 31, 2020	2019	TV-MA	107 min

14. Top 10 Most Featured Actors in Indian Netflix Movies

```
188 SELECT
189 UNNEST(STRING_TO_ARRAY(casts, ',')) as actor
190 COUNT(*) as total_content
191 FROM netflix
192 WHERE country ILIKE '%india'
193 GROUP BY 1
194 ORDER BY 2 DESC
195 LIMIT 10
```

Data Output Messages Notifications

	actors text	total_content bigint
1	Arjun Rampal	25
2	Paresh Rawal	25
3	Shah Rukh Khan	24
4	Om Puri	23
5	Boman Irani	23
6	Akshay Kumar	22
7	Amitabh Bachchan	20

15. Classifying Content as 'Good' or 'Bad' Based on Descriptions Containing 'Kill' or 'Violence'

```
174 SELECT
175     category,
176     TYPE,
177     COUNT(*) AS content_count
178 FROM (
179     SELECT
180         *,
181         CASE
182             WHEN description ILIKE '%kill%' OR description ILIKE '%violence%' THEN 'Bad'
183             ELSE 'Good'
184         END AS category
185     FROM netflix
186 ) AS categorized_content
187 GROUP BY 1,2
188 ORDER BY 2
```

Data Output			Messages	Notifications
	category	total_content		
	text	bigint		
1	Good Content	8465		
2	Bad_Content	342		

Key Beneficial Insights from this Analysis:

1. Movies Dominate the Catalog

→ Prioritize short-form content for global appeal and faster production cycles.

2. India & U.S. Lead in Content Volume

→ Strengthen partnerships and investment in these high-yield regions.

3. Drama & Comedy Are Top Genres

→ Focus on these genres for broader audience engagement and retention.

4. Top Actors/Directors Drive Rewatchability

→ Use talent analytics to inform casting and promotional strategies.

5. Recent Years Show Sharp Content Growth

→ Indicates aggressive expansion; timing content drops can boost subscriber spikes.